3.4 Algebraic Topology — Solutions

Solutions to problems.md. Standard results are quoted from Hatcher (or notes.md) by name. Throughout, $p : \mathbb{R} \to S^1$, $p(s) = e^{2\pi i s}$.

Warm-up

1. $\pi_1(S^n) = 0$ **for** $n \geq 2$

Let N, S be the north and south poles, $U = S^n \setminus \{N\}$, $V = S^n \setminus \{S\}$. Both are open, and stereographic projection gives $U \cong V \cong \mathbb{R}^n$, so both are simply connected. $U \cap V = S^n \setminus \{N, S\} \cong S^{n-1} \times (-1, 1)$, which is path-connected because $n - 1 \geq 1$. Choose a basepoint in $U \cap V$. By van Kampen, $\pi_1(U) * \pi_1(V) \to \pi_1(S^n)$ is surjective, and the source is trivial. S^n is path-connected, so it is simply connected. □

(For $n = 1$, $U \cap V$ has two components and the argument fails, as it must.)

2. No retraction $D^2 \to S^1$; Brouwer in dimension 2

Suppose $r : D^2 \to S^1$ with $r \circ i = \mathrm{id}_{S^1}$, where i is the inclusion. Then $r_* \circ i_* = \mathrm{id}$ on $\pi_1(S^1, 1) \cong \mathbb{Z}$. But i_* factors through $\pi_1(D^2, 1) = 0$ (D^2 is convex, so contractible). So the identity of $\mathbb{Z}$ is the zero map, a contradiction.

Now let $h : D^2 \to D^2$ have no fixed point. For each x, the ray starting at $h(x)$ and passing through x meets S^1 in exactly one point $r(x)$ beyond x (or at x). Explicitly $r(x) = x + t(x)\,(x - h(x))$ where $t(x) \geq 0$ is the non-negative root of $|x + t(x - h(x))|^2 = 1$; the quadratic formula shows t depends continuously on x because $x - h(x) \neq 0$. If $x \in S^1$ then $t(x) = 0$ and $r(x) = x$. So r is a retraction $D^2 \to S^1$, which is impossible. □

3. Torus vs. wedge of circles

(a) $T^2 = S^1 \times S^1$ and π_1 of a product is the product, so $\pi_1(T^2) \cong \mathbb{Z}^2$. For $S^1 \vee S^1$, take U and V to be the wedge minus a point of the second, respectively first, circle. Each deformation retracts onto one circle, and $U \cap V$ is contractible (a cross shape around the wedge point). Van Kampen gives $\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F_2$.

(b) A homotopy equivalence induces an isomorphism on π_1 (both spaces are path-connected, so basepoints don't matter up to isomorphism). $\mathbb{Z}^2$ is abelian and F_2 is not ($ab \neq ba$ as reduced words). So they are not homotopy equivalent. □

4. $\tilde{H}_*(S^n)$ **by Mayer–Vietoris**

Claim: $\tilde{H}_k(S^n) \cong \mathbb{Z}$ if $k = n$, and 0 otherwise, for $n \geq 0$.

Base case $n = 0$: S^0 is two points, so $H_0 = \mathbb{Z}^2$, $\tilde{H}_0 = \mathbb{Z}$, and higher groups vanish.

Inductive step, $n \geq 1$: with U, V as in Problem 1, both are contractible and $U \cap V \simeq S^{n-1}$ is non-empty. The reduced Mayer–Vietoris sequence contains

$$\tilde{H}_k(U) \oplus \tilde{H}_k(V) \to \tilde{H}_k(S^n) \xrightarrow{\partial} \tilde{H}_{k-1}(U \cap V) \to \tilde{H}_{k-1}(U) \oplus \tilde{H}_{k-1}(V).$$

The outer terms vanish, so $\tilde{H}_k(S^n) \cong \tilde{H}_{k-1}(S^{n-1})$ for every k. Induction finishes the proof. □

5. Euler characteristics of surfaces

M_g has one 0-cell, $2g$ 1-cells and one 2-cell: $\chi(M_g) = 1 - 2g + 1 = 2 - 2g$. N_h ($h \geq 1$) has one 0-cell, h 1-cells and one 2-cell: $\chi(N_h) = 2 - h$.

They agree exactly when $h = 2g$ (with $g \geq 1$), e.g. $\chi(T^2) = \chi(K) = 0$.

H_1 from the cellular chain complex (or as the abelianization of π_1): with one 0-cell, $d_1 = 0$, so $H_1 = \mathbb{Z}^{\#\text{1-cells}}/\operatorname{im} d_2$, and d_2 sends the 2-cell to the exponent sums of its attaching word.

- For M_g the word $[a_1, b_1] \cdots [a_g, b_g]$ has all exponent sums 0, so $H_1(M_g) \cong \mathbb{Z}^{2g}$, torsion-free.
- For N_h the word $a_1^2 \cdots a_h^2$ gives $d_2(e^2) = 2(a_1 + \cdots + a_h)$. Since $a_1, \ldots, a_{h-1}, a_1 + \cdots + a_h$ is a basis of $\mathbb{Z}^h$, $H_1(N_h) \cong \mathbb{Z}^{h-1} \oplus \mathbb{Z}/2$.

$H_1(N_h)$ has torsion and $H_1(M_g)$ does not, so $M_g \not\simeq N_h$ for all g, h. □

6. Contractible spaces

Let $F : X \times I \to X$ be a homotopy from id_X to the constant map c at a point x_1. For any x, $t \mapsto F(x, t)$ is a path from x to x_1, so X is path-connected.

Homology: $\mathrm{id}_* = c_*$ on $\tilde{H}_n(X)$ by homotopy invariance. But c factors as $X \to \{x_1\} \hookrightarrow X$, and $\tilde{H}_n(\mathrm{pt}) = 0$ for all n. So $\mathrm{id}_* = 0$ and $\tilde{H}_n(X) = 0$.

π_1: the constant map $X \to \{x_1\}$ is a homotopy equivalence (with inverse the inclusion), so it induces an isomorphism $\pi_1(X, x_0) \cong \pi_1(\mathrm{pt}) = 0$ for any x_0 (a homotopy equivalence induces isomorphisms on π_1 at every basepoint, by Hatcher Prop. 1.18). □

Core

7. $\mathbb{RP}^2$

(a) $\mathbb{Z}/2 = \{\pm 1\}$ acts on S^2 by $x \mapsto -x$. For $x \in S^2$, the open set $U = \{y : |y - x| < 1\}$ satisfies $U \cap (-U) = \emptyset$ (if $y \in U \cap (-U)$, then $2 = |x - (-x)| \leq |x - y| + |y + x| < 2$). So this is a covering space action, and $q : S^2 \to S^2/\pm 1 = \mathbb{RP}^2$ is a 2-sheeted normal covering with deck group $\mathbb{Z}/2$. Since S^2 is simply connected (Problem 1), it is the universal cover, so $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2$.

(b) Let $f : \mathbb{RP}^2 \to S^1$. Then $f_* : \mathbb{Z}/2 \to \mathbb{Z}$ is zero because $\mathbb{Z}$ is torsion-free. $\mathbb{RP}^2$ is path-connected and locally path-connected, so by the lifting criterion f lifts to $\tilde{f} : \mathbb{RP}^2 \to \mathbb{R}$ with $p\tilde{f} = f$. $\mathbb{R}$ is contractible: $\tilde{f}_t = (1 - t)\tilde{f}$ is a homotopy to 0, and $p \circ \tilde{f}_t$ is a homotopy from f to a constant. □

8. Klein bottle

(a) Corners: $(0, 0) \sim (1, 0)$ via the first relation; $(0, 0) \sim (1, 1)$ and $(1, 0) \sim (0, 1)$ via the second. So all four corners become one 0-cell v. Let b be the vertical edge $\{0\} \times I = \{1\} \times I$, oriented upward, and a the bottom edge, oriented left to right. Because $(x, 1) \sim (1 - x, 0)$, the top edge read left to right is a^{-1}. Reading the boundary of the square counterclockwise from $(0, 0)$: bottom a, right side b, top from right to left a, left side downward b^{-1}. So the 2-cell is attached along $abab^{-1}$. K has one 0-cell, two 1-cells, one 2-cell; the 1-skeleton is $S^1 \vee S^1$, and attaching a 2-cell kills the normal closure of its word:

$$\pi_1(K) \cong \langle a, b \mid abab^{-1} \rangle.$$

(b) The relation says $bab^{-1} = a^{-1}$. In S_3 let $r = (1\,2\,3)$ and $s = (1\,2)$; then $srs^{-1} = r^{-1}$. So $a \mapsto r$, $b \mapsto s$ defines a homomorphism $\pi_1(K) \to S_3$. Since $rs \neq sr$, also $ab \neq ba$ in $\pi_1(K)$. So $\pi_1(K)$ is non-abelian.

(c) $H_1(K) \cong \pi_1(K)^{ab} = \langle a, b \mid 2a = 0 \rangle$ (in additive notation the relator is $a + b + a - b = 2a$) $\cong \mathbb{Z} \oplus \mathbb{Z}/2$. □

9. $H_*(\mathbb{RP}^n)$

$\mathbb{RP}^n$ has one cell e^k in each dimension $0 \leq k \leq n$, and $\mathbb{RP}^k = \mathbb{RP}^{k-1} \cup e^k$ with attaching map the quotient $\varphi : S^{k-1} \to \mathbb{RP}^{k-1}$.

The cellular boundary $d_k(e^k) = d \cdot e^{k-1}$, where d is the degree of $S^{k-1} \xrightarrow{\varphi} \mathbb{RP}^{k-1} \xrightarrow{q} \mathbb{RP}^{k-1}/\mathbb{RP}^{k-2} \cong S^{k-1}$.

Compute d by local degrees. A point y in the open $(k-1)$-cell has exactly two preimages $x, -x$ under $q\varphi$. On a small neighborhood of x, $q\varphi$ is a homeomorphism onto a neighborhood of y; call its local degree $\epsilon = \pm 1$. Since $q\varphi(-z) = q\varphi(z)$, near $-x$ the map is $(q\varphi|_{\text{near } x}) \circ a$ where a is the antipodal map of S^{k-1}, of degree $(-1)^k$. So the local degree at $-x$ is $\epsilon(-1)^k$ and

$$d = \epsilon(1 + (-1)^k) = \begin{cases} \pm 2, & k \text{ even} \\ 0, & k \text{ odd.} \end{cases}$$

(For $k = 1$: $d_1 = 0$ since there is only one 0-cell, consistent.)

The cellular chain complex is $\mathbb{Z}$ in each degree $0..n$ with $d_k = 0$ for k odd and $d_k = \pm 2$ for k even, $k \geq 2$. Hence:

- $H_0 = \mathbb{Z}$;
- for odd $k < n$: $\ker d_k = \mathbb{Z}$, $\operatorname{im} d_{k+1} = 2\mathbb{Z}$, so $H_k = \mathbb{Z}/2$;
- for even k with $0 < k \leq n$: d_k is injective, so $H_k = 0$;
- for $k = n$ odd: $\ker d_n = \mathbb{Z}$ and nothing above, so $H_n = \mathbb{Z}$;
- $H_k = 0$ for $k > n$. □

10. Antipodal map and hairy ball

(a) First, a reflection $\rho(x_0, x_1, \ldots, x_n) = (-x_0, x_1, \ldots, x_n)$ has degree -1. Give S^n the CW structure with two 0-cells, two 1-cells, ..., two n-cells, built from S^{n-1} by attaching the two hemispheres $\{x_0 \geq 0\}$ and $\{x_0 \leq 0\}$ as e^n_+, e^n_- (inductively the S^{n-1} is the equator $x_0 = 0$; any CW structure on S^{n-1} works). In cellular chains $d(e^n_+) = d(e^n_-)$ (both attached by the identity of the equator), so $H_n(S^n)$ is generated by $e^n_+ - e^n_-$. ρ is cellular, fixes the equator, and swaps e^n_+ and e^n_- (preserving their identification with D^n that restricts to the identity on the boundary). So $\rho_*(e^n_+ - e^n_-) = e^n_- - e^n_+$, degree -1.

Any reflection in a hyperplane through the origin is conjugate by a rotation (homotopic to the identity, since $SO(n+1)$ is path-connected) to ρ, so has degree -1. The antipodal map is the composite of the $n+1$ coordinate reflections, so $\deg a = (-1)^{n+1}$.

(b) If $n = 2k - 1$ is odd, $v(x_1, \ldots, x_{2k}) = (-x_2, x_1, -x_4, x_3, \ldots, -x_{2k}, x_{2k-1})$ is a unit tangent vector field ($v(x) \cdot x = 0$, $|v(x)| = |x| = 1$).

Conversely suppose v is a nowhere-zero tangent field on S^n; replacing v by $v/|v|$ assume $|v(x)| = 1$ and $v(x) \perp x$. Then $f_t(x) = \cos(\pi t)\, x + \sin(\pi t)\, v(x)$ has $|f_t(x)|^2 = \cos^2 + \sin^2 = 1$, so it is a homotopy in S^n from id to a. Then $1 = \deg a = (-1)^{n+1}$, so n is odd. □

11. Brouwer via homology

Suppose $r : D^n \to S^{n-1}$ is a retraction, $n \geq 1$. Then $r \circ i = \mathrm{id}$ and on $\tilde{H}_{n-1}$ the identity of $\tilde{H}_{n-1}(S^{n-1}) \cong \mathbb{Z}$ factors through $\tilde{H}_{n-1}(D^n) = 0$ (contractible). Contradiction. If $h : D^n \to D^n$ had no fixed point, the ray construction of Problem 2 (same formula, same continuity argument) would give such a retraction. So h has a fixed point. □

12. Two-sheeted covers of $S^1 \vee S^1$

Connected 2-sheeted covers correspond (up to isomorphism, ignoring basepoints) to conjugacy classes of index-2 subgroups of $F_2 = \langle a, b \rangle$. Index-2 subgroups are normal, so we need the index-2 subgroups themselves. These are kernels of surjections $F_2 \to \mathbb{Z}/2$, which are determined by the images of a, b in $\mathbb{Z}/2$, not both zero: three choices. So there are exactly **three** connected 2-sheeted covers, all **normal**.

In each case the cover is a graph with vertices v_0, v_1 and, at each vertex, one outgoing and one incoming edge of each label.

1. $a \mapsto 0$, $b \mapsto 1$: an a-loop at each vertex, and two b-edges $v_0 \to v_1$, $v_1 \to v_0$. Taking the b-edge $v_0 \to v_1$ as a maximal tree, the subgroup (at v_0) is free on a, bab^{-1}, b^2.
2. $a \mapsto 1$, $b \mapsto 0$: symmetric; subgroup $\langle b,\ aba^{-1},\ a^2 \rangle$.
3. $a \mapsto 1$, $b \mapsto 1$: two a-edges and two b-edges, each pair forming a 2-cycle between v_0 and v_1. With tree the a-edge $v_0 \to v_1$, the subgroup (words of even length) is free on a^2, ba^{-1}, ab.

Each subgroup has rank $3 = 2(2-1)+1$, consistent with Problem 19. □

13. Lens spaces

Free action. Suppose $k \cdot (z, w) = (z, w)$ for some $(z, w) \in S^3$. If $z \neq 0$, $\zeta^k = 1$ so $m \mid k$. If $z = 0$ then $w \neq 0$, so $\zeta^{kq} = 1$, $m \mid kq$, and $\gcd(m, q) = 1$ gives $m \mid k$. So only the identity has fixed points.

Covering space action. A free action of a finite group G on a Hausdorff space is a covering space action: given y, the points gy ($g \in G$) are distinct, so choose pairwise disjoint open $U_g \ni gy$ and set $U = \bigcap_g g^{-1}U_g$, an open neighborhood of y. Then $gU \subseteq U_g$, which is disjoint from $U \subseteq U_e$ for $g \neq e$.

So $S^3 \to L(m, q)$ is a normal covering with deck group $\mathbb{Z}/m$. S^3 is simply connected (Problem 1), path-connected and locally path-connected, so $\pi_1(L(m, q)) \cong \mathbb{Z}/m$. □

14. $H_*(K)$ by Mayer–Vietoris

Cut K along a circle to write $K = A \cup B$ with A, B open Möbius bands (slightly thickened) and $A \cap B$ an open annulus, which deformation retracts to a circle C. Then $A \simeq B \simeq S^1$ (core circles), $A \cap B \simeq S^1$, and the inclusion $C \hookrightarrow A$ sends the generator of $H_1(C)$ to ± 2 times the generator of $H_1(A)$, since the boundary of a Möbius band winds twice around its core. Same for B.

Reduced Mayer–Vietoris:

$$0 = H_2(A) \oplus H_2(B) \to H_2(K) \to H_1(A \cap B) \xrightarrow{\Phi} H_1(A) \oplus H_1(B) \to H_1(K) \to \tilde{H}_0(A \cap B) = 0.$$

With suitable generators, $\Phi(1) = (2, -2)$ (signs depend on orientation choices but Φ is injective in any case). Hence $H_2(K) = \ker \Phi = 0$, and

$$H_1(K) \cong \mathbb{Z}^2 / \langle (2, -2) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2,$$

using the basis $(1,-1),(0,1)$ of $\mathbb{Z}^2$, in which the subgroup is $2\cdot(1,-1)$. $H_0(K)=\mathbb{Z}$ and $H_k(K)=0$ for $k\geq 3$ (all terms vanish). This agrees with Problem 8(c). $\square$

15. Invariance of dimension

$\mathbb{R}^0$ is a point, not homeomorphic to $\mathbb{R}^n$ for $n\geq 1$. Let $m,n\geq 1$ and $h:\mathbb{R}^m\to\mathbb{R}^n$ a homeomorphism. Then h restricts to a homeomorphism of pairs $(\mathbb{R}^m,\mathbb{R}^m\setminus\{0\})\to(\mathbb{R}^n,\mathbb{R}^n\setminus\{h(0)\})$, so their relative homology groups are isomorphic.

For any $x\in\mathbb{R}^n$, the long exact sequence of the pair in reduced homology, with $\tilde{H}_*(\mathbb{R}^n)=0$, gives $H_k(\mathbb{R}^n,\mathbb{R}^n\setminus\{x\})\cong\tilde{H}_{k-1}(\mathbb{R}^n\setminus\{x\})\cong\tilde{H}_{k-1}(S^{n-1})$, which is $\mathbb{Z}$ for $k=n$ and 0 otherwise. Comparing gives $m=n$. $\square$

Qualifying-exam level

16. Retracts split homology

(a) $r\circ i=\mathrm{id}_A$ gives $r_*i_*=\mathrm{id}$, so i_* is injective in every degree. In the long exact sequence of (X,A),

$$H_n(A)\xrightarrow{i_*}H_n(X)\xrightarrow{j_*}H_n(X,A)\xrightarrow{\partial}H_{n-1}(A)\xrightarrow{i_*}H_{n-1}(X),$$

injectivity of the last i_* forces $\partial=0$, so j_* is surjective, and we get short exact sequences $0\to H_n(A)\to H_n(X)\to H_n(X,A)\to 0$. They split because r_* is a left inverse of i_*. Hence $H_n(X)\cong H_n(A)\oplus H_n(X,A)$.

(b) Let M be the Möbius band and $C=\partial M\cong S^1$. M deformation retracts onto its core, so $H_1(M)\cong\mathbb{Z}$, and $i_*:H_1(C)\to H_1(M)$ is multiplication by ± 2. If $r:M\to C$ were a retraction, then $r_*\circ i_*=\mathrm{id}_{\mathbb{Z}}$, i.e. $\pm 2\,r_*(1)=1$ in $\mathbb{Z}$, impossible. (Note i_* *is* injective here; the obstruction is that its image $2\mathbb{Z}$ is not a direct summand, contradicting (a).) $\square$

17. Maps between S^2 and T^2

(a) The universal cover of T^2 is $\mathbb{R}^2\to T^2$. S^2 is simply connected and locally path-connected, so by the lifting criterion any $f:S^2\to T^2$ lifts to $\tilde{f}:S^2\to\mathbb{R}^2$. The straight-line homotopy $(1-t)\tilde{f}$ contracts $\tilde{f}$ to a constant in $\mathbb{R}^2$, and composing with the covering map gives a nullhomotopy of f.

(b) Let $A=S^1\vee S^1\subset T^2$ be the 1-skeleton of the standard CW structure, and $g:T^2\to T^2/A\cong S^2$ the quotient (the quotient of a 2-cell by its boundary is S^2). (T^2,A) is a good pair, so $\tilde{H}_2(T^2/A)\cong H_2(T^2,A)$ and g_* is identified with $j_*:H_2(T^2)\to H_2(T^2,A)$. In the long exact sequence

$$0=H_2(A)\to H_2(T^2)\xrightarrow{j_*}H_2(T^2,A)\xrightarrow{\partial}H_1(A)\xrightarrow{i_*}H_1(T^2),$$

$i_*:\mathbb{Z}^2\to\mathbb{Z}^2$ is an isomorphism (both are the abelianization of $F_2\to\mathbb{Z}^2$ at the level of generators a,b). So $\partial=0$ and j_* is an isomorphism $\mathbb{Z}\to\mathbb{Z}$. So g has degree ± 1 on H_2.

(c) By (a), f is nullhomotopic, so $g\circ f$ is nullhomotopic and has degree 0. Equivalently $f_*=0$ on H_2, so $(gf)_*=g_*f_*=0$. $\square$

18. $\mathbb{CP}^2$ **vs.** $S^2 \vee S^4$

Homology. $\mathbb{CP}^2$ has cells in dimensions $0, 2, 4$, so all cellular boundaries vanish and $H_k = \mathbb{Z}$ for $k = 0, 2, 4$, zero otherwise. For a wedge of CW complexes, $\tilde{H}_k(S^2 \vee S^4) \cong \tilde{H}_k(S^2) \oplus \tilde{H}_k(S^4)$, again $\mathbb{Z}$ in degrees $2, 4$. So the homology groups agree (and, by the universal coefficient theorem, so do the cohomology groups: everything is free).

Cup products. Let $W = S^2 \vee S^4$, $i : S^2 \hookrightarrow W$ and $r : W \to S^2$ the map collapsing S^4 to the wedge point, so $r \circ i = \mathrm{id}$. Then $i^* r^* = \mathrm{id}$ on $H^2(S^2) \cong \mathbb{Z}$. Since $H^2(W) \cong \mathbb{Z}$, $r^* : \mathbb{Z} \to \mathbb{Z}$ has a left inverse, so it is an isomorphism. Let u generate $H^2(S^2)$ and $y = r^* u$, a generator of $H^2(W)$. Then

$$y \smile y = r^*(u \smile u) = 0,$$

because $u \smile u \in H^4(S^2) = 0$.

Suppose $h : \mathbb{CP}^2 \to W$ is a homotopy equivalence. Then $h^* : H^*(W) \to H^*(\mathbb{CP}^2)$ is a ring isomorphism. So $h^*(y) = \pm x$, and $h^*(y \smile y) = x \smile x = x^2$, which generates $H^4(\mathbb{CP}^2) \cong \mathbb{Z}$, in particular is nonzero. But $h^*(y \smile y) = h^*(0) = 0$. Contradiction. □

19. Subgroups of free groups

(a) Let X be a connected graph (1-dimensional CW complex) and $p : \tilde{X} \to X$ a covering. Let $\tilde{X}^0 = p^{-1}(X^0)$. Each open edge e of X is contractible, hence simply connected and locally path-connected, so the inclusion $e \hookrightarrow X$ lifts and $p^{-1}(e)$ is a disjoint union of copies of e, each mapped homeomorphically. Each characteristic map $[0, 1] \to X$ of an edge lifts (path lifting) to paths in $\tilde{X}$ whose endpoints lie in $\tilde{X}^0$, one for each point of the fibre over its starting point. These lifts, with $\tilde{X}^0$ as vertices, give $\tilde{X}$ the structure of a graph (the topology agrees because p is a local homeomorphism and X has the weak topology).

Let F be free on a set S. Then $F \cong \pi_1(X)$ for $X = \bigvee_{s \in S} S^1$. Given $H \leq F$, the classification theorem gives a connected cover $\tilde{X}_H \to X$ with $p_* \pi_1(\tilde{X}_H) = H$, and p_* is injective, so $H \cong \pi_1(\tilde{X}_H)$. $\tilde{X}_H$ is a connected graph; collapsing a maximal tree T (contractible subcomplex) is a homotopy equivalence onto a wedge of circles, one per edge not in T. So H is free.

(b) Let $X = \bigvee_{i=1}^{n} S^1$ (one vertex, n edges), so $\chi(X) = 1 - n$. The cover $\tilde{X}_H$ has k sheets, so it has k vertices and kn edges: $\chi(\tilde{X}_H) = k(1-n)$. A connected graph is homotopy equivalent to a wedge of r circles, where r is the rank of its π_1, and χ is a homotopy invariant (equal to $\operatorname{rank} H_0 - \operatorname{rank} H_1 = 1 - r$). So $1 - r = k(1-n)$ and

$$r = k(n-1) + 1.$$

(c) Any index-2 subgroup of F_2 (e.g. the kernel of $F_2 \to \mathbb{Z}/2$, $a \mapsto 1$, $b \mapsto 0$) has rank $2(2-1)+1 = 3$, so it is isomorphic to F_3. □

20. Fixed points and antipodal points

(a) If $f(x) \neq x$ for all x, then $(1-t)f(x) - tx \neq 0$ for all $t \in I$: if it were 0, taking norms gives $1 - t = t$, so $t = \frac{1}{2}$ and $f(x) = x$, excluded. So

$$f_t(x) = \frac{(1-t)f(x) - tx}{|(1-t)f(x) - tx|}$$

is a homotopy from f to the antipodal map a, and $\deg f = \deg a = (-1)^{n+1}$ (Problem 10a).

(b) If $f(x) \neq -x$ for all x, the same argument with $(1-t)f(x)+tx$ (which vanishes only if $t = \frac{1}{2}$ and $f(x) = -x$) gives $f \simeq \mathrm{id}$, so $\deg f = 1$.

(c) Let n be even and suppose $f(x) \neq x$ and $f(x) \neq -x$ for all x. By (a), $\deg f = (-1)^{n+1} = -1$; by (b), $\deg f = 1$. Contradiction.

Application. Let $n = 2k \geq 2$ and $F : \mathbb{RP}^n \to \mathbb{RP}^n$, with $q : S^n \to \mathbb{RP}^n$ the universal cover. $F \circ q : S^n \to \mathbb{RP}^n$ lifts through q because S^n is simply connected and locally path-connected; call the lift $f : S^n \to S^n$, so $qf = Fq$. By (c) there is x with $f(x) = \pm x$. Then $F(q(x)) = q(f(x)) = q(\pm x) = q(x)$, a fixed point. □

21. Borsuk–Ulam for $n = 2$

Suppose $g : S^2 \to S^1$ satisfies $g(-x) = -g(x)$. Let $\eta : I \to S^2$ be a path from x_0 to $-x_0$ (half of a great circle). Then $\gamma = \eta \cdot (-\eta)$, where $(-\eta)(s) = -\eta(s)$ runs from $-x_0$ back to x_0, is a loop at x_0.

Let $\tilde{h}$ be the lift of $g \circ \eta$ to $\mathbb{R}$ starting at $\tilde{s}_0$, where $p(\tilde{s}_0) = g(x_0)$. It ends at a lift of $g(-x_0) = -g(x_0) = p(\tilde{s}_0 + \frac{1}{2})$, so $\tilde{h}(1) = \tilde{s}_0 + \frac{q}{2}$ for some **odd** integer q.

Now $g \circ (-\eta) = -(g \circ \eta)$, and since $-p(s) = p(s + \frac{1}{2})$, every function $\tilde{h} + \frac{1}{2} + j$ ($j \in \mathbb{Z}$) is a lift of $g \circ (-\eta)$. Choose $j = \frac{q-1}{2}$ so that it starts at $\tilde{s}_0 + \frac{q}{2} = \tilde{h}(1)$; it ends at $\tilde{h}(1) + \frac{q}{2} = \tilde{s}_0 + q$. So the lift of $g \circ \gamma$ starting at $\tilde{s}_0$ ends at $\tilde{s}_0 + q$, meaning $[g \circ \gamma] = q \neq 0$ in $\pi_1(S^1) \cong \mathbb{Z}$.

But S^2 is simply connected, so γ is nullhomotopic and so is $g \circ \gamma$: $[g \circ \gamma] = 0$. Contradiction.

Borsuk–Ulam. If $f : S^2 \to \mathbb{R}^2$ had $f(x) \neq f(-x)$ for all x, then

$$g(x) = \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

would be a continuous map $S^2 \to S^1$ with $g(-x) = -g(x)$, which we have just shown is impossible. □